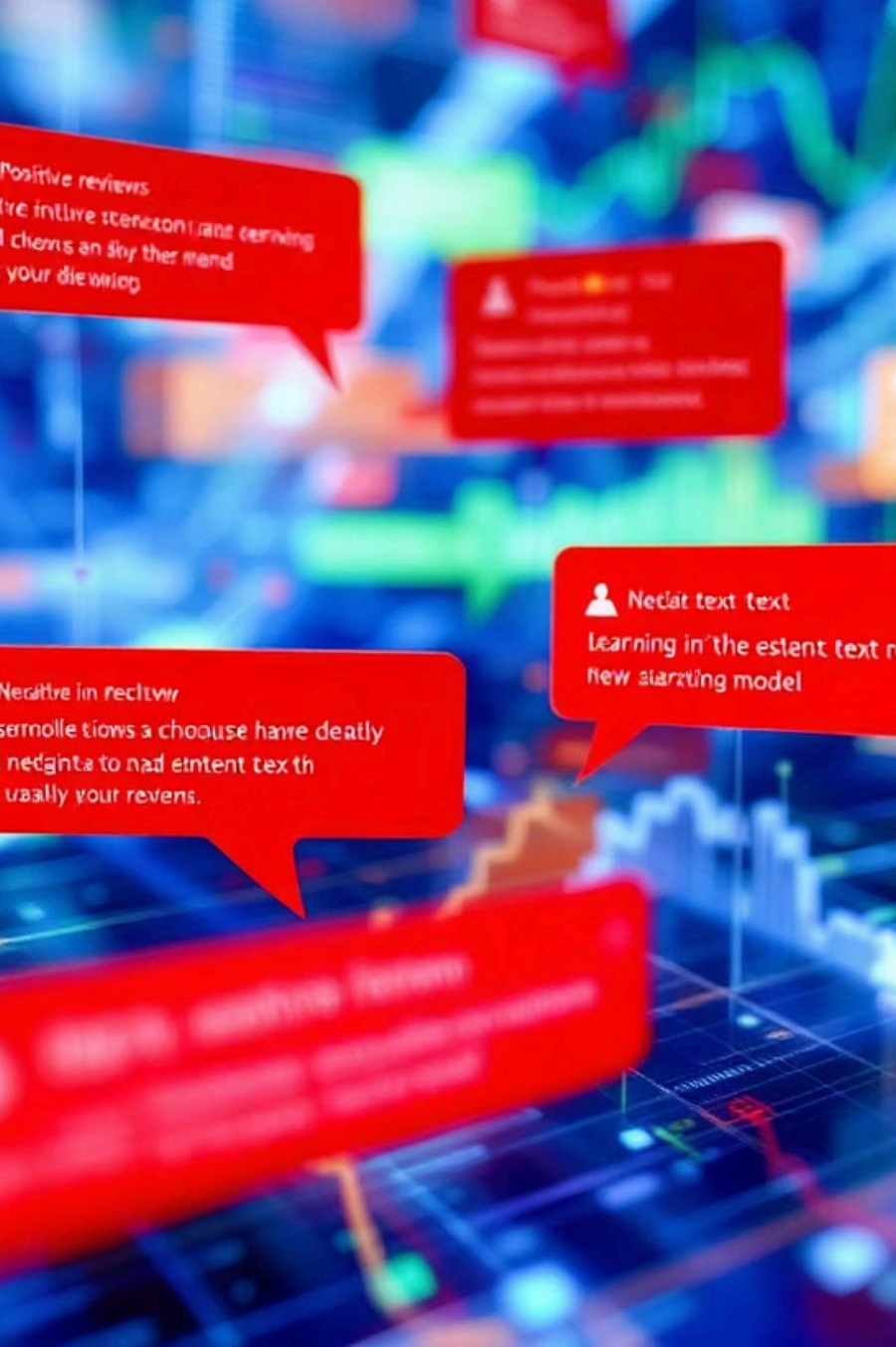




Sentiment Prediction of Text Review

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Semester 5 Project

Domain: Natural Language Processing (NLP) in Marketing

Motivation & Importance



Reduces manual workload

for review analysis.



Protects brand reputation

by quickly identifying negative feedback.



Enables data-driven decisions

for product/service improvements.



Enhances customer experience

by surfacing pain points.

Methodology Workflow



Data Ingestion

Collect review data.



Data Cleaning & Preprocessing

Remove noise, handle missing values, tokenize text, remove stopwords.



Feature Engineering

- Extract word frequencies
- Use TF-IDF, word embeddings, or semantic features



Model Training

- Experiment with models: Naive Bayes, SVM, RNN, CNN



Evaluation & Optimization

- Use metrics: Accuracy, F1 Score
- Hyperparameter tuning



Deployment (if applicable)

Integrate model into business workflow.

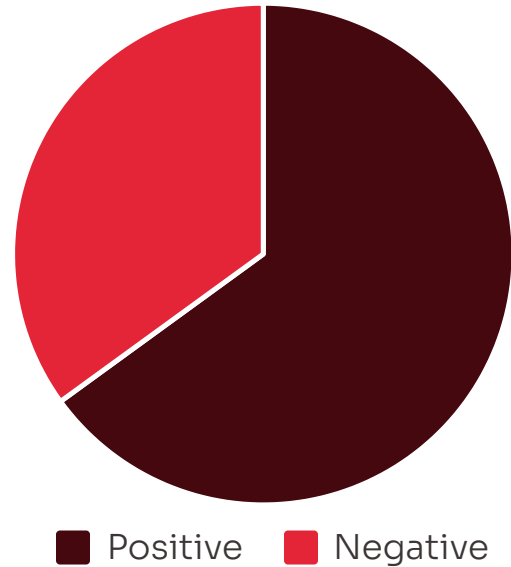
Tools & Technologies

Tool/Library	Purpose
Python	Programming language
Scikit-Learn	Machine learning models
TensorFlow	Deep learning (if needed)
NLTK	Text preprocessing
Pandas/NumPy	Data manipulation

Example Visualizations

Visualizations help understand sentiment patterns and distribution:

- **Pie Chart/Bar Graph:** Distribution of positive vs. negative reviews
- **Word Cloud:** Common words in positive and negative reviews



Conclusion & Future Scope

Conclusion:

- Successfully automated sentiment detection for text reviews.
- Provided actionable insights for business improvement.

Future Scope:

- Implement deep learning models for higher accuracy.
- Multilingual sentiment analysis.
- Real-time review processing.
- Interactive dashboards for visualization.



Thank You!

For your time and attention.

Kapish Shukla



Thank You

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